

Tool Code Library

Test Case Document

Project	Tool Code Library
Document Type	Test Case Specification
Framework	Django 5.x / Python 3.12
Database	Microsoft SQL Server
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Introduction

Purpose

This document defines the test cases for the **Tool Code Library** web application built using the Django framework. The application manages tool code creation, approval workflows, master data administration, and report generation for a manufacturing organisation.

Scope

The test cases cover all major functional modules:

- Authentication (Login / Logout)
- Role-based access control (dynamic role system)
- Tool Code creation with dynamic form generation
- Approval workflow (Approve / Reject / Resubmit)
- Master data management (Field, Attributes, Suppliers, etc.)
- User and Role management
- Reports and PDF download with dynamic footers
- Draft save functionality
- Audit logging

Test Case ID Convention

Prefix	Module
TC-AUTH	Authentication
TC-NAV	Navigation / Role-based access
TC-CC	Tool Code Creation
TC-APR	Approval Workflow
TC-FLD	Field Master
TC-FAT	Field Attributes
TC-FFV	Field Fixed Values
TC-SUP	Supplier Master
TC-RM	Raw Material Master
TC-JT	Joint Type Master
TC-FMT	Format Master (PDF Footer)
TC-USR	User Management
TC-ROLE	Role Master
TC-PWD	Change Password
TC-RPT	Reports
TC-PDF	PDF Generation
TC-DFT	Draft Functionality
TC-AUD	Audit Log

Result Legend

- **Pass** — Test passed; actual result matches expected result.
- **Fail** — Test failed; defect to be logged.

- *Pending* — Test not yet executed.

Authentication

Table 1: TC-AUTH-001 — Successful Login

Field	Detail
Test Case ID	TC-AUTH-001
Title	Successful Login with Valid Credentials
Priority	High
Preconditions	A user account exists in the system with an assigned active role.
Test Steps	1. Navigate to <code>http://127.0.0.1:8000/</code>. 2. Enter a valid username. 3. Enter the correct password. 4. Click Login.
Expected Result	User is redirected to the Dashboard. The navbar displays menu items appropriate to the user's role.
Actual Result	The user was successfully redirected to the Dashboard, and the navbar displayed the correct role-specific menu items.
Result	Pass

Table 2: TC-AUTH-002 — Login with Invalid Password

Field	Detail
Test Case ID	TC-AUTH-002
Title	Login with Incorrect Password
Priority	High
Preconditions	User account exists.
Test Steps	1. Navigate to the login page. 2. Enter a valid username. 3. Enter an incorrect password. 4. Click Login.
Expected Result	The login page reloads and displays the error message: <i>“Invalid username or password.”</i> . No redirect occurs.
Actual Result	The login page reloaded and successfully displayed the “Invalid username or password.” error message. The user remained on the login screen.
Result	Pass

Table 3: TC-AUTH-003 — Login with Deleted Role

Field	Detail
Test Case ID	TC-AUTH-003
Title	Login Blocked When User's Role is Deactivated
Priority	High
Preconditions	A user exists whose assigned role has been deactivated (<code>is_deleted = True</code>).
Test Steps	1. Navigate to the login page. 2. Enter credentials of the user with the deactivated role. 3. Click Login.
Expected Result	Login is blocked. Error message: <i>"Your assigned role is currently inactive."</i> . An audit log entry is written with level WARNING .
Actual Result	Login was blocked with the correct error message, and a WARNING level audit log entry was verified in the database.
Result	Pass

Table 4: TC-AUTH-004 — Logout

Field	Detail
Test Case ID	TC-AUTH-004
Title	Successful Logout Terminates Session
Priority	High
Preconditions	User is logged in.
Test Steps	1. Click Logout in the navbar. 2. Confirm the SweetAlert2 confirmation dialog. 3. After redirect, attempt to navigate to <code>/dashboard/</code> directly.
Expected Result	User is redirected to the login page. Attempting to access <code>/dashboard/</code> redirects back to login — session is fully terminated.
Actual Result	The session was successfully terminated, redirecting to the login page, and direct URL access to the dashboard was blocked.
Result	Pass

Role-Based Navigation

Table 5: TC-NAV-001 — User Role Sidebar

Field	Detail
Test Case ID	TC-NAV-001
Title	Sidebar Shows Correct Tabs for a User Role
Priority	High
Preconditions	A role exists with only <code>can_access_tool_code</code> and <code>can_create_requests</code> enabled. A user is assigned this role.
Test Steps	1. Log in as the User-role user. 2. Observe the sidebar.
Expected Result	Sidebar shows: Dashboard, Tool Codes, Change Password. Master Data, User Creation, Reports, and Role Master are not visible.
Actual Result	The sidebar correctly displayed only the Dashboard, Tool Codes, and Change Password tabs as per role permissions.
Result	Pass

Table 6: TC-NAV-002 — Approver Role Sidebar

Field	Detail
Test Case ID	TC-NAV-002
Title	Approver Role Can Access Tool Code and Reports
Priority	High
Preconditions	A role has <code>can_approve_requests</code> and <code>can_access_reports</code> enabled.
Test Steps	1. Log in as an Approver-role user. 2. Observe the sidebar and the Tool Codes table.
Expected Result	Sidebar shows Tool Codes, Reports, Change Password. The Tool Codes table shows all non-draft requests. Approve/Reject action buttons are visible for Pending rows.
Actual Result	The Approver successfully saw the Tool Codes and Reports tabs, and the action buttons were correctly visible for Pending rows.
Result	Pass

Table 7: TC-NAV-003 — Direct URL Access Without Permission

Field	Detail
Test Case ID	TC-NAV-003
Title	Unauthorised Direct URL Access Redirects to Dashboard
Priority	High
Preconditions	Logged in as a User-role user (no master data access).
Test Steps	1. Manually type <code>/supplier-master/</code> in the browser address bar and press Enter. 2. Repeat for <code>/user-creation/</code> , <code>/field-attributes/</code> , <code>/role-master/</code> .
Expected Result	All URLs redirect to <code>/dashboard/</code> without displaying the restricted page. No error is thrown.
Actual Result	All unauthorized URLs successfully redirected to the dashboard without displaying restricted content or throwing errors.
Result	Pass

Tool Code Creation

Table 8: TC-CC-001 — Dynamic Form Loads on Field Selection

Field	Detail
Test Case ID	TC-CC-001
Title	Selecting a Field Dynamically Loads Its Sub-fields
Priority	High
Preconditions	A Field (e.g. Boring Tool Holder) exists with FieldAttribute rows configured (Bore Size, No. of Teeth, O/A Length).
Test Steps	1. Open Tool Codes page and click + Create Code. 2. From the Primary Description dropdown, select <i>Boring Tool Holder</i>. 3. Observe the form below the supplier section.
Expected Result	A details section appears with inputs for Bore Size, No. of Teeth, and O/A Length dynamically loaded via AJAX from <code>/get-field-attributes/</code> . Joint Type and Raw Material dropdowns are also shown.
Actual Result	The form dynamically expanded to show Bore Size, No. of Teeth, and O/A Length inputs successfully loaded via AJAX.
Result	Pass

Table 9: TC-CC-002 — Dropdown Sub-field Populated from FieldAttributeOption

Field	Detail
Test Case ID	TC-CC-002
Title	Select-type Sub-field Shows Correct Dropdown Options
Priority	High
Preconditions	A Field has a sub-field of type <code>select</code> with options (e.g. Drill Type: TWIST DRILL, STEP DRILL).
Test Steps	1. Open Create Code modal. 2. Select the field that has the dropdown sub-field. 3. Click the dropdown sub-field.
Expected Result	The dropdown shows exactly the options configured in Field Attributes Master (e.g. TWIST DRILL, STEP DRILL). Options from <code>FieldAttributeOption</code> table are used, not from <code>FieldFixedValue</code> .
Actual Result	The dropdown correctly populated with the specific options configured in the Field Attributes Master.
Result	Pass

Table 10: TC-CC-003 — Tool Code Generation Format

Field	Detail
Test Case ID	TC-CC-003
Title	Submitted Request Generates Correct Tool Code Format
Priority	High
Preconditions	Boring Tool Holder field exists with short_code=BTH. Attributes: Bore Size=125, No. of Teeth=40, O/A Length=200. Joint Type first letter=S. Raw Material=STEEL. Fixed Value=A.
Test Steps	1. Fill in the Create Code form with the above values. 2. Click Submit Request. 3. Observe the tool code in the Tool Codes table.
Expected Result	Tool code generated is BTH-A-125X40-200-S-ST (first two attrs joined with X, third with dash, joint first letter, material first two letters). Code is unique.
Actual Result	The system successfully generated the unique, correctly formatted tool code BTH-A-125X40-200-S-ST.
Result	Pass

Table 11: TC-CC-004 — Validation: Required Fields Missing

Field	Detail
Test Case ID	TC-CC-004
Title	Submitting Without Required Fields Shows Validation Error
Priority	High
Preconditions	User is on the Create Code modal.
Test Steps	1. Select a Primary Description. 2. Leave Supplier 1, Joint Type, and Raw Material empty. 3. Click Submit Request.
Expected Result	SweetAlert2 error popup is shown: <i>“Please fill in all the required fields.”</i> . No ToolRequest record is created.
Actual Result	Submission was successfully blocked by the SweetAlert2 popup, and no incomplete record was saved to the database.
Result	Pass

Table 12: TC-CC-005 — Validation: Duplicate Suppliers

Field	Detail
Test Case ID	TC-CC-005
Title	Same Supplier in Supplier 1 and Supplier 2 is Rejected
Priority	Medium
Preconditions	At least one supplier exists.
Test Steps	1. Open Create Code modal. 2. Select the same supplier in both Supplier 1 and Supplier 2. 3. Fill all other required fields. 4. Click Submit Request.
Expected Result	SweetAlert2 warning: <i>“Supplier 1 and Supplier 2 cannot be the exact same company.”</i> . No record saved.
Actual Result	The system correctly detected identical suppliers and blocked the submission with the appropriate warning message.
Result	Pass

Table 13: TC-CC-006 — Supplier Code Auto-Fill

Field	Detail
Test Case ID	TC-CC-006
Title	Selecting a Supplier Auto-fills the Ordering Code
Priority	Medium
Preconditions	A supplier with ordering code S001 exists.
Test Steps	1. Open Create Code modal. 2. Select the supplier from Supplier 1 dropdown.
Expected Result	Supplier Ordering Code 1 field is automatically populated with S001 without any user interaction.
Actual Result	The Supplier Ordering Code 1 field automatically populated with S001 upon selecting the supplier.
Result	Pass

Table 14: TC-CC-007 — Tool Code Uniqueness

Field	Detail
Test Case ID	TC-CC-007
Title	Duplicate Tool Code Gets a Numeric Suffix
Priority	High

Field	Detail
Preconditions	A ToolRequest with tool code BTH-A-125X40-200-S-ST already exists.
Test Steps	1. Submit a second request with identical field values.
Expected Result	The new request is assigned tool code BTH-A-125X40-200-S-ST-01 (or -02 etc.). Both records exist in the database.
Actual Result	The duplicate request was successfully saved with the appropriate numeric suffix (-01) appended to the tool code.
Result	Pass

Draft Functionality

Table 15: TC-DFT-001 — Save as Draft with Minimum Data

Field	Detail
Test Case ID	TC-DFT-001
Title	Saving a Draft Requires Only Primary Description
Priority	Medium
Preconditions	User has <code>can.create.requests</code> permission.
Test Steps	1. Open Create Code modal. 2. Select a Primary Description only. Leave all other fields empty. 3. Click Save as Draft.
Expected Result	A <code>ToolRequest</code> record is created with <code>status = Draft</code> and <code>tool_code = [DRAFT]</code> . Success message is shown.
Actual Result	The draft was successfully saved with minimal data, retaining the <code>[DRAFT]</code> placeholder and Draft status.
Result	Pass

Table 16: TC-DFT-002 — Draft Not Visible to Other Users

Field	Detail
Test Case ID	TC-DFT-002
Title	Draft Requests Are Hidden from Other Users Including Approvers
Priority	High
Preconditions	User A has a Draft request. User B has Approver role.
Test Steps	1. Log in as User A and save a draft. 2. Log in as User B (Approver). 3. View the Tool Codes table.
Expected Result	User A's draft does not appear in User B's Tool Codes table. Approvers only see non-draft requests from all users.
Actual Result	The draft request was completely hidden from the Approver's view, confirming data isolation.
Result	Pass

Table 17: TC-DFT-003 — Edit and Submit a Draft

Field	Detail
Test Case ID	TC-DFT-003
Title	A Draft Can Be Opened, Completed, and Submitted
Priority	High
Preconditions	User has a draft request in the Tool Codes table.
Test Steps	1. Click the Edit (pencil) button on the Draft row. 2. Fill all required fields. 3. Click Submit Request.
Expected Result	Status changes from Draft to Pending. A proper tool code is generated and saved. The request becomes visible to Approvers.
Actual Result	The draft was successfully opened, fully populated, and submitted, generating a valid tool code and moving to Pending status.
Result	Pass

Approval Workflow

Table 18: TC-APR-001 — Approve a Pending Request

Field	Detail
Test Case ID	TC-APR-001
Title	Approver Can Approve a Pending Tool Request
Priority	High
Preconditions	A Pending <code>ToolRequest</code> exists. User has <code>can_approve_requests</code> permission.
Test Steps	1. Log in as an Approver. 2. On the Tool Codes table, locate a Pending row. 3. Click the green Approve (thumbs up) button. 4. Confirm the SweetAlert2 dialog.
Expected Result	Request status changes to Approved . <code>reviewed_by</code> is set to the Approver. <code>reviewed_at</code> is set to current timestamp. Success flash message is displayed.
Actual Result	The request was successfully approved, the status updated correctly, and the review timestamps/user data were recorded.
Result	Pass

Table 19: TC-APR-002 — Reject with Reason

Field	Detail
Test Case ID	TC-APR-002
Title	Approver Can Reject a Request with a Mandatory Reason
Priority	High
Preconditions	A Pending <code>ToolRequest</code> exists.
Test Steps	1. Click the red Reject (thumbs down) button on a Pending row. 2. The Reject modal opens. 3. Enter a rejection reason. 4. Click Confirm Reject.
Expected Result	Status changes to Rejected . <code>reject_reason</code> is saved. The requester's row now shows an Edit (pencil) button.
Actual Result	The request was successfully rejected with the mandatory reason saved, and the requester's row updated to allow editing.
Result	Pass

Table 20: TC-APR-003 — Reject Without Reason is Blocked

Field	Detail
Test Case ID	TC-APR-003
Title	Rejection Reason Field is Mandatory
Priority	Medium
Preconditions	Reject modal is open.
Test Steps	1. Open the Reject modal. 2. Leave the reason textarea empty. 3. Click Confirm Reject.
Expected Result	Browser HTML5 validation prevents submission. The rejection is not processed.
Actual Result	HTML5 validation successfully prevented the rejection submission when the reason field was left blank.
Result	Pass

Table 21: TC-APR-004 — User Edits and Resubmits Rejected Request

Field	Detail
Test Case ID	TC-APR-004
Title	Requester Can Edit and Resubmit a Rejected Request
Priority	High
Preconditions	A <code>ToolRequest</code> with status Rejected exists for the logged-in user.
Test Steps	1. Log in as the original requester. 2. Click the Edit button on the Rejected row. 3. Verify the rejection reason banner is displayed in the modal. 4. Modify one or more fields. 5. Click Submit Request.
Expected Result	Status resets to Pending. A new tool code is generated. Old <code>reject_reason</code> is cleared. Old attributes are replaced.
Actual Result	The edited request was successfully resubmitted, the rejection reason cleared, and the status returned to Pending with a newly generated code.
Result	Pass

Table 22: TC-APR-005 — Review Page Displays All Submitted Data

Field	Detail
Test Case ID	TC-APR-005
Title	Review Request Page Shows All Dynamic Attributes
Priority	High
Preconditions	A submitted request exists with dynamic attributes (e.g. Drill Diameter, Flute Length, Drill Type).
Test Steps	1. Click View on any request row. 2. Observe the Specifications section.
Expected Result	All <code>ToolRequestAttribute</code> rows are displayed with their names and values. Dropdown-selected values (e.g. TWIST DRILL) display the text, not the database ID.
Actual Result	The review modal successfully displayed all dynamic <code>ToolRequestAttribute</code> rows with their correct names and text values.
Result	Pass

Field Master

Table 23: TC-FLD-001 — Create a New Field

Field	Detail
Test Case ID	TC-FLD-001
Title	Admin Can Create a New Tool Type (Field)
Priority	High
Preconditions	Logged in as a user with <code>can_access_master</code> .
Test Steps	1. Navigate to Master → Field Form. 2. Enter Field Name, Short Code, Field Size Type, Field Size, Field Message. 3. Click Save.
Expected Result	The new field appears in the saved fields table. It is available in the Primary Description dropdown on the Create Code form.
Actual Result	The new field was successfully saved and immediately became available in the Create Code primary description dropdown.
Result	Pass

Table 24: TC-FLD-002 — Deactivate a Field

Field	Detail
Test Case ID	TC-FLD-002
Title	Deactivating a Field Hides It from Create Code Form
Priority	Medium
Preconditions	An active Field exists.
Test Steps	1. On the Field Form page, toggle the field's status to inactive. 2. Navigate to Create Code and open the Primary Description dropdown.
Expected Result	The deactivated field does not appear in the dropdown. Existing tool requests using this field are not affected.
Actual Result	The deactivated field was successfully hidden from the dropdown without affecting existing requests.
Result	Pass

Field Attributes Master

Table 25: TC-FAT-001 — Add Number Attribute

Field	Detail
Test Case ID	TC-FAT-001
Title	Admin Can Add a Number Sub-field to a Field
Priority	High
Preconditions	A Field exists. Logged in with master access.
Test Steps	1. Navigate to Master → Field Attributes. 2. Select the Field from the dropdown. 3. Enter Attribute Name = <i>Bore Size</i>, Input Type = Number, Order = 1, Required = On. 4. Click Add.
Expected Result	The attribute appears in the configured sub-fields table with badge <i>Number</i> and Required = Yes.
Actual Result	The Number attribute was successfully added and correctly configured as required in the sub-fields table.
Result	Pass

Table 26: TC-FAT-002 — Add Dropdown Attribute with Options

Field	Detail
Test Case ID	TC-FAT-002
Title	Selecting Dropdown Type Expands Option Builder Inline
Priority	High
Preconditions	Field Attributes page is open for a selected field.
Test Steps	1. Enter Attribute Name = <i>Drill Type</i>. 2. Change Input Type to <i>Select (Dropdown)</i>. 3. Observe the Dropdown Options builder expanding. 4. Type <i>TWIST DRILL</i> and press Enter. Type <i>STEP DRILL</i> and press Enter. 5. Click Add.
Expected Result	Attribute is created with Input Type = Dropdown. The table shows a sub-row with pills: TWIST DRILL, STEP DRILL. Both options exist in the <code>FieldAttributeOption</code> table.
Actual Result	The dropdown option builder expanded successfully, and the entered options were saved properly to the database.
Result	Pass

Table 27: TC-FAT-003 — Edit Dropdown Attribute via Modal

Field	Detail
Test Case ID	TC-FAT-003
Title	Edit Modal for Dropdown Shows Existing Options
Priority	Medium
Preconditions	A Dropdown-type attribute with options exists.
Test Steps	1. Click Edit on the Dropdown attribute row. 2. Observe the Dropdown Edit modal. 3. Click the × button on one existing option. 4. Type a new option and click + Add Option.
Expected Result	The modal opens showing existing option pills. The removed option is soft-deleted (<code>is_deleted = True</code>). The new option is added. Changes are reflected after the modal closes and page reloads.
Actual Result	The edit modal correctly allowed soft-deleting existing options and adding new ones, reflecting accurately after reload.
Result	Pass

Supplier, Raw Material, and Joint Type Masters

Table 28: TC-SUP-001 — Add and Deactivate Supplier

Field	Detail
Test Case ID	TC-SUP-001
Title	Admin Can Add a Supplier and Deactivate It
Priority	Medium
Preconditions	Logged in with master access.
Test Steps	1. Navigate to Master → Supplier. 2. Enter Supplier Name and Ordering Code; click Save . 3. Toggle the supplier's status to inactive.
Expected Result	Supplier is added and shown in the table. After deactivation, it no longer appears in the Supplier dropdowns on the Create Code form. The record is not deleted from the database.
Actual Result	The supplier was added successfully and later deactivated, appropriately disappearing from forms while remaining in the database.
Result	Pass

Table 29: TC-RM-001 — Add Raw Material

Field	Detail
Test Case ID	TC-RM-001
Title	Admin Can Add a Raw Material
Priority	Medium
Preconditions	Logged in with master access.
Test Steps	1. Navigate to Master → Raw Material. 2. Enter a material name (e.g. <i>Carbide</i>). 3. Click Save .
Expected Result	Material appears in the table and in the Raw Material dropdown on the Create Code form.
Actual Result	The raw material was successfully saved and correctly appeared in the Create Code form dropdown.
Result	Pass

Table 30: TC-JT-001 — Add Joint Type

Field	Detail
Test Case ID	TC-JT-001
Title	Admin Can Add a Joint Type
Priority	Medium
Preconditions	Logged in with master access.
Test Steps	1. Navigate to Master → Joint Type. 2. Enter a joint type name (e.g. <i>Welded Joint</i>). 3. Click Save.
Expected Result	Joint type appears in the table and in the Joint Type dropdown on the Create Code form. Its first letter is used in the generated tool code.
Actual Result	The joint type was added successfully, appeared in the dropdown, and its first letter correctly applied to tool codes.
Result	Pass

Format Master (PDF Footer)

Table 31: TC-FMT-001 — Save Footer Configuration

Field	Detail
Test Case ID	TC-FMT-001
Title	Admin Can Configure All Three PDF Footers
Priority	Medium
Preconditions	Logged in with master access.
Test Steps	1. Navigate to Master → Format Master. 2. Enter text in Footer Left, Footer Centre, and Footer Right fields. 3. Observe the live footer preview strip update in real-time. 4. Click Save Footer.
Expected Result	Success flash message. The PdfFooterConfig table is updated. The live preview reflects the typed text before saving.
Actual Result	The live preview successfully updated during typing, and the footer configuration was saved correctly.
Result	Pass

Table 32: TC-FMT-002 — Footer Reflected in Downloaded PDF

Field	Detail
Test Case ID	TC-FMT-002
Title	Saved Footer Text Appears on Downloaded PDF
Priority	High
Preconditions	Footer configuration saved. A completed ToolRequest exists.
Test Steps	1. Navigate to Reports. 2. Click the PDF download button on any row. 3. Open the downloaded PDF and inspect the footer area.
Expected Result	The footer shows the saved Left text at bottom-left, Centre text at bottom-centre, and Right text at bottom-right, separated by a horizontal rule. Page number appears at the top-right corner.
Actual Result	The downloaded PDF correctly displayed the saved footer text and the page number in their designated locations.
Result	Pass

Table 33: TC-FMT-003 — Footer Changes Apply to Future Downloads

Field	Detail
Test Case ID	TC-FMT-003
Title	Changing Footer Config Updates All Subsequent PDF Downloads
Priority	Medium
Preconditions	An existing footer config is saved.
Test Steps	1. Change the Footer Left text to a new value and save. 2. Download a PDF from Reports.
Expected Result	The downloaded PDF shows the new footer text. No server restart or code change is required.
Actual Result	The updated footer configuration immediately applied to new PDF downloads without requiring a server restart.
Result	Pass

PDF Generation

Table 34: TC-PDF-001 — PDF Contains All Request Data

Field	Detail
Test Case ID	TC-PDF-001
Title	Downloaded PDF Contains All Submitted Information
Priority	High
Preconditions	A completed <code>ToolRequest</code> with all fields filled exists.
Test Steps	1. Click the PDF button on the Reports page for the request. 2. Open and inspect the downloaded PDF.
Expected Result	PDF contains: Tool Code, Description, Fixed Value, Joint Type, Raw Material, all dynamic attributes (e.g. Bore Size, Diameter, Drill Type), Supplier 1 and 2, ordering codes, remarks, audit trail (requested by / on, reviewed by / on).
Actual Result	The generated PDF accurately included all submitted tool specifications, master data references, and audit trail information.
Result	Pass

Table 35: TC-PDF-002 — Page Number at Top Right

Field	Detail
Test Case ID	TC-PDF-002
Title	Page Number Appears at Top Right on Every Page
Priority	Medium
Preconditions	A multi-page tool request PDF can be generated (many attributes).
Test Steps	1. Download a PDF with enough content to span two pages. 2. Inspect the top-right corner of each page.
Expected Result	<i>Page 1</i> , <i>Page 2</i> etc. appear at the top-right of each page. No page number appears in the footer area.
Actual Result	Multi-page PDFs correctly displayed the page numbers at the top right of each respective page.
Result	Pass

Table 36: TC-PDF-003 — User Cannot Download Another User’s Request

Field	Detail
Test Case ID	TC-PDF-003
Title	User Cannot Download PDF of Another User’s Request
Priority	High
Preconditions	User A and User B each have a ToolRequest.
Test Steps	1. Log in as User A. 2. Manually navigate to /download-pdf/<User B’s request ID>/.
Expected Result	User A is redirected to the Tool Codes page with an error message: “ <i>You do not have permission to download this request.</i> ”.
Actual Result	Attempting to access another user’s PDF successfully blocked access and redirected with the correct error message.
Result	Pass

User Management

Table 37: TC-USR-001 — Create New User

Field	Detail
Test Case ID	TC-USR-001
Title	Admin Can Create a New User with a Dynamic Role
Priority	High
Preconditions	Logged in with <code>can_manage_users</code> permission. At least one active Role exists.
Test Steps	1. Navigate to User Creation. 2. Fill in Full Name, Username, Email, Password. 3. Select a Role from the dynamic dropdown. 4. Click Create User.
Expected Result	A <code>User</code> and <code>UserProfile</code> record are created. The new user can log in immediately with the assigned permissions.
Actual Result	The user and associated UserProfile were created successfully, and the dynamic role assignment worked flawlessly.
Result	Pass

Table 38: TC-USR-002 — Duplicate Username Rejected

Field	Detail
Test Case ID	TC-USR-002
Title	Creating a User with an Existing Username Shows an Error
Priority	Medium
Preconditions	A user with username <i>john</i> exists.
Test Steps	1. Navigate to User Creation. 2. Enter username <i>john</i> and fill other fields. 3. Click Create User.
Expected Result	Error message: “ <i>Username already exists.</i> ”. No new user record is created.
Actual Result	The system correctly identified the duplicate username and prevented the creation of the new account, displaying the proper error.
Result	Pass

Role Master

Table 39: TC-ROLE-001 — Create a New Role

Field	Detail
Test Case ID	TC-ROLE-001
Title	Admin Can Create a Role with Custom Permissions
Priority	High
Preconditions	Logged in with <code>can_manage_roles</code> permission.
Test Steps	1. Navigate to Role Master. 2. Enter a Role Name (e.g. <i>Senior Approver</i>). 3. Toggle on <code>can_access_tool_code</code>, <code>can_approve_requests</code>, <code>can_access_reports</code>. 4. Click Create Role.
Expected Result	Role is created and appears in the roles table. It is available in the Role dropdown on the User Creation page. Users assigned this role see only the permitted tabs.
Actual Result	The role was created successfully with custom permissions, and users assigned this role only saw permitted tabs.
Result	Pass

Table 40: TC-ROLE-002 — Deactivate a Role

Field	Detail
Test Case ID	TC-ROLE-002
Title	Deactivating a Role Blocks Users Assigned to It
Priority	High
Preconditions	A role exists with users assigned to it.
Test Steps	1. On the Role Master page, toggle the role status to inactive. 2. Attempt to log in as a user with this role.
Expected Result	Login is blocked with message: <i>“Your assigned role is currently inactive.”</i> . An audit warning log entry is created.
Actual Result	Deactivating the role successfully blocked logins for assigned users and generated the correct warning log.
Result	Pass

Table 41: TC-ROLE-003 — Super Admin Role Cannot Be Deactivated

Field	Detail
Test Case ID	TC-ROLE-003
Title	Super Admin Role is Protected from Deactivation
Priority	High
Preconditions	A role named <i>Super Admin</i> exists.
Test Steps	1. On the Role Master page, attempt to toggle the Super Admin role inactive.
Expected Result	Action is blocked. Error message: <i>“Cannot deactivate the Super Admin role.”</i> . Role remains active.
Actual Result	The system successfully protected the Super Admin role, preventing its deactivation and displaying the expected error.
Result	Pass

Change Password

Table 42: TC-PWD-001 — Successful Password Change

Field	Detail
Test Case ID	TC-PWD-001
Title	User Can Change Password and Remains Logged In
Priority	High
Preconditions	User is logged in.
Test Steps	1. Navigate to Change Password. 2. Enter the correct old password. 3. Enter a new password of 8+ characters. 4. Confirm the new password. 5. Click Update Password.
Expected Result	Password is updated. Session remains active (<code>update_session_auth_hash</code> called). Success message displayed. New password works on next login.
Actual Result	The password was successfully updated, the session remained active, and the new password worked for subsequent logins.
Result	Pass

Table 43: TC-PWD-002 — Wrong Old Password Rejected

Field	Detail
Test Case ID	TC-PWD-002
Title	Entering Incorrect Old Password Shows an Error
Priority	Medium
Preconditions	User is logged in.
Test Steps	1. Navigate to Change Password. 2. Enter an incorrect old password. 3. Click Update Password.
Expected Result	Error message: “ <i>Old password is incorrect.</i> ”. Password is not changed.
Actual Result	The system correctly detected the wrong old password and prevented the update, showing the appropriate error.
Result	Pass

Table 44: TC-PWD-003 — Password Mismatch Rejected

Field	Detail
Test Case ID	TC-PWD-003
Title	New Password and Confirm Password Must Match
Priority	Medium
Preconditions	User is logged in.
Test Steps	1. Enter correct old password. 2. Enter new password = <i>abc12345</i>. 3. Enter confirm password = <i>abc12346</i> (different). 4. Click Update Password.
Expected Result	Error message: “ <i>New passwords do not match.</i> ”. Password unchanged.
Actual Result	The mismatched confirmation password was successfully detected, preventing the update and displaying the error.
Result	Pass

Reports

Table 45: TC-RPT-001 — Date Filter Narrows Results

Field	Detail
Test Case ID	TC-RPT-001
Title	Date Range Filter Returns Only Matching Requests
Priority	Medium
Preconditions	Multiple ToolRequest records exist across different dates.
Test Steps	1. Navigate to Reports. 2. Set From Date and To Date to a specific range. 3. Click Apply Date Filter.
Expected Result	Only requests created within the selected date range are shown. Record count updates accordingly.
Actual Result	The date filter successfully restricted the table to display only requests matching the specified date range.
Result	Pass

Table 46: TC-RPT-002 — Status Filter

Field	Detail
Test Case ID	TC-RPT-002
Title	Status Filter Buttons Show Only Matching Records
Priority	Medium
Preconditions	Requests of all statuses exist.
Test Steps	1. Click Approved filter button. 2. Observe the table. 3. Click Rejected. 4. Click All.
Expected Result	Each filter shows only matching rows. The selected button becomes dark/active. Clicking All restores the full table.
Actual Result	The status filter buttons successfully toggled the table to display only the relevant rows, with the 'All' button restoring the full view.
Result	Pass

Table 47: TC-RPT-003 — Tool Code Search

Field	Detail
Test Case ID	TC-RPT-003
Title	Search Box Filters Table by Tool Code in Real Time
Priority	Medium
Preconditions	Multiple requests with different tool codes exist.
Test Steps	1. Type part of a tool code (e.g. <i>BTH</i>) into the search box.
Expected Result	Only rows whose tool code contains <i>BTH</i> are shown. The record count and pagination update immediately without a page reload.
Actual Result	The real-time search immediately filtered the rows by the entered tool code without requiring a page reload.
Result	Pass

Audit Log

Table 48: TC-AUD-001 — Successful Login Written to Audit Log

Field	Detail
Test Case ID	TC-AUD-001
Title	Successful Login Creates INFO Audit Log Entry
Priority	Medium
Preconditions	User account exists.
Test Steps	1. Log in successfully. 2. Open Django Admin → AuditLog.
Expected Result	An <code>AuditLog</code> entry exists with <code>level = INFO</code> , the correct <code>user</code> , <code>path = /</code> , <code>method = POST</code> , and message containing “logged in successfully”.
Actual Result	The INFO level audit log entry was successfully verified in the database with the correct user, path, and method details.
Result	Pass

Table 49: TC-AUD-002 — Failed Login Written to Audit Log

Field	Detail
Test Case ID	TC-AUD-002
Title	Failed Login Attempt Creates ERROR Audit Log Entry
Priority	Medium
Preconditions	None.
Test Steps	1. Attempt to log in with a non-existent username. 2. Open Django Admin → AuditLog.
Expected Result	An <code>AuditLog</code> entry exists with <code>level = ERROR</code> , <code>user = null</code> , the attempted username in the message, and the IP address recorded.
Actual Result	The failed login attempt correctly generated an ERROR level audit log entry capturing the attempted username and IP.
Result	Pass

Test Execution Summary

Module	Total Cases	Result
Authentication	4	Pass
Role-Based Navigation	3	Pass
Tool Code Creation	7	Pass
Draft Functionality	3	Pass
Approval Workflow	5	Pass
Field Master	2	Pass
Field Attributes Master	3	Pass
Supplier / RM / Joint Type	3	Pass
Format Master	3	Pass
PDF Generation	3	Pass
User Management	2	Pass
Role Master	3	Pass
Change Password	3	Pass
Reports	3	Pass
Audit Log	2	Pass
Total	49	

Status	Count
Pass	49
Fail	0
Pending	0
Total	49

Prepared By:

Reviewed By:

Date:

Signature

Signature